

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Fuel System Clean Care Kit, Part Number 4919073
- Compuchek™ fitting, Part Number 3100221
- Orificed diagnostic fuel line, Part Number 3164621
- Five leak test isolation tools, Part Number 4919546
- Fuel pump service tool, Part Number 5299753, from kit, Part Number 5299721
- Tester, fuel system leak, Part Number 4918354
- Fuel drain line, Part Number 3164673
- Hose, Part Number 3164673
- Graduated beaker, Part Number 4919139
- INSITE™ electronic service tool
- Service tool, Part Number 5299753
- Fuel return hose, Cummins® Part Number 3164618
- Graduated cylinder, Cummins® Part Number 4919139
- Banjo flow adapter fitting, Cummins® Part Number 316461.

Additional Service Items

- No additional service items required.

General Information

The following procedures are used to diagnose fuel system issues. These checks and measurements are referenced throughout the applicable troubleshooting and fault code trees, as needed.

This procedure is **not** intended to take the place of the troubleshooting tree repair direction.

Reference the appropriate troubleshooting symptom tree for repair direction.

Note : The fuel line from the engine control module (ECM) cooling plate to the high-pressure fuel pump contains a filter screen at the high-pressure fuel pump banjo fitting. The screen must be examined and cleaned or the banjo bolt replaced if the screen is damaged.

Preparatory Steps

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ WARNING ⚠

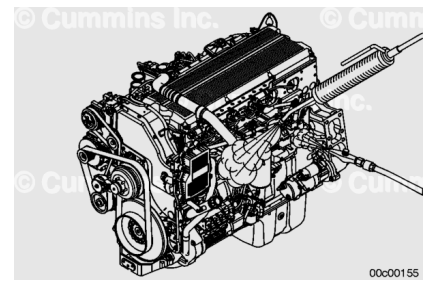
When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ CAUTION ⚠

Clean all fittings before disassembly. Dirt or contaminants can damage the fuel system.

⚠ CAUTION ⚠

Hold banjo fittings while tightening the banjo bolt to prevent fitting rotation. Allowing the banjo fitting to rotate may damage the fuel line.



Before servicing any fuel system components, (such as fuel lines, fuel pump, injectors, etc.) which will expose the fuel system or internal engine component to potential contaminants prior to disassembly, clean the fittings, mounting hardware, and the area around the component to be removed. Dirt or contaminants can be introduced into the fuel system and engine if the surrounding areas are **not** cleaned, resulting in damage to the fuel system and engine. Refer to Procedure 000-009 in Section 0.

(/qs3/pubsys2/xml/en/procedures/99/99-000-009.html)

To prevent engine damage from debris or contamination, cover, cap, or plug any openings as soon as possible when servicing the fuel system. Caps and plugs can be found in Fuel System Clean Care Kit, Part Number 4919073.

⚠ WARNING ⚠

High-pressure fuel spray can cause serious injury or death.

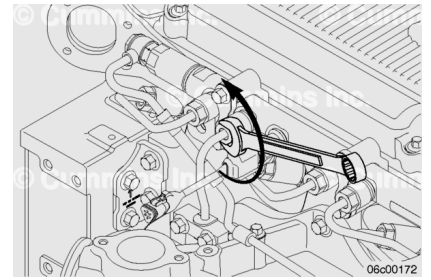
⚠ WARNING ⚠

Depending on the circumstance, diesel fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death, or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.

⚠ WARNING ⚠

Do not bleed the fuel system of a hot engine; this can result in fuel spilling onto a hot exhaust manifold, which can cause a fire.

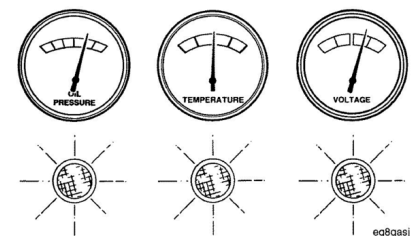
Before servicing the fuel system, relieve any residual pressure in the fuel system. Refer to Procedure 006-081 in Section 6. (</qs3/pubsys2/xml/en/procedures/377/377-006-081.html>)



Low-Pressure System Check

General Information

The low-pressure system check consists of a number of measurements and checks to make sure that the low-pressure fuel system is functioning properly. These checks will vary, depending on whether or **not** the engine will start.

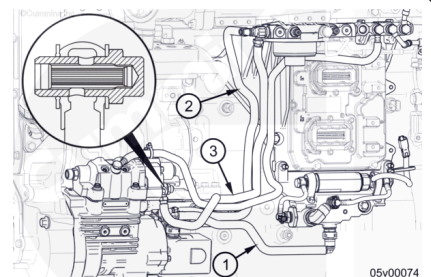


With Quick Disconnect Fuel Lines:

Quick disconnect nylon fuel line configurations will **no** longer require a fuel-cooled, ECM cooling plate.

Fuel supply lines equipped with quick disconnect fuel line fittings can be found at the following locations:

- Fuel lift pump support outlet to fuel pump gear pump inlet (1).

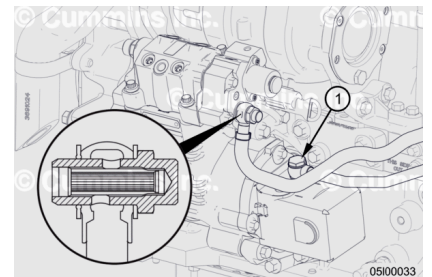


- Fuel pump gear pump outlet to fuel filter inlet (2).
- Fuel filter outlet to high-pressure fuel pump (3). The fuel line from the filter outlet to the high-pressure fuel pump contains a filter screen at the high-pressure fuel pump banjo fitting. The screen **must** be examined and cleaned, or the banjo bolt replaced if the screen is damaged.

Without Quick Disconnect Fuel Lines:

The fuel line from the gear pump to the filter head (1) is a steel line without quick disconnect.

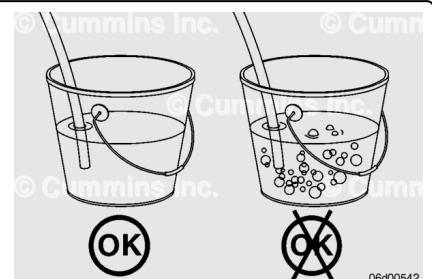
The fuel line from the filter head to the high-pressure fuel pump contains a filter screen at the high-pressure fuel pump banjo fitting. The screen **must** be examined and cleaned, or the banjo bolt replaced, if the screen is damaged.



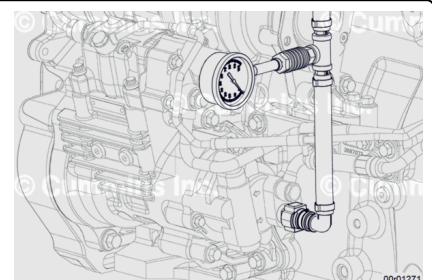
Measurement - Engine Will Start

Verify the fuel system has been primed. Refer to Procedure 005-234 in Section 5. (</qs3/pubsys2/xml/en/procedures/377/377-005-234-tr.html>)

Check for air in the fuel. Refer to Procedure 006-003 in Section 6. (</qs3/pubsys2/xml/en/procedures/377/377-006-003.html>)

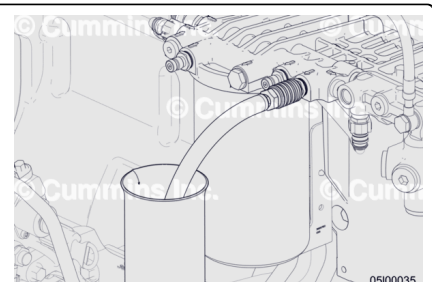


Measure the fuel inlet restriction. Refer to Procedure 006-020 in Section 6. (</qs3/pubsys2/xml/en/procedures/377/377-006-020.html>)



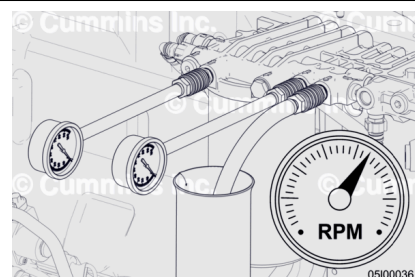
Measure Fuel Pressure

- Remove the threaded plug from the outlet side of the fuel filter head and install a Compuchek™ fitting, Part Number 3100221, or equivalent.



- Install the orificed diagnostic fuel line, Part Number 3164621, at the Compuchek™ fitting.

- Install a 0 to 2068 kPa [0 to 300 psi] pressure gauge at the Compuchek™ fitting at the inlet to the fuel filter head.
- Install a 0 to 2068 kPa [0 to 300 psi] pressure gauge at the Compuchek™ fitting at the outlet to the fuel filter head.
- Wait for the lift pump cycle to complete (approximately 60 seconds).
- Operate the engine at high idle and observe the filter inlet pressure.
- Operate the engine at high idle and observe the filter outlet pressure.



Measurements

	kpa	psi
Fuel Filter Restriction (max)	138	20

If the difference between the filter inlet pressure and filter outlet pressure is greater than the specification, replace the fuel filter.

Operate the engine at high idle and observe the fuel gear pump pressure at the fuel filter inlet.

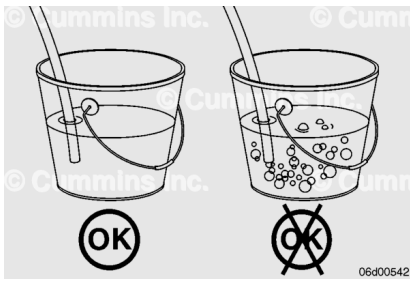
Measurements

	kpa	psi
Gear Pump Pressure at High Idle (min)	1000	145

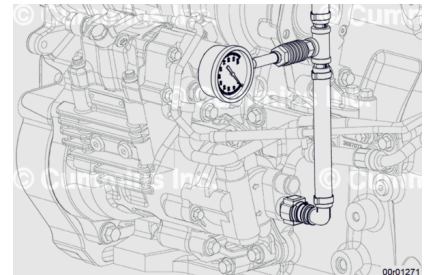
Measurement - Engine Will Not Start

Verify the fuel system has been primed. Refer to Procedure 005-234 in Section 5. (</qs3/pubsys2/xml/en/procedures/377/377-005-234-tr.html>)

Check for air in the fuel. Refer to Procedure 006-003 in Section 6. (</qs3/pubsys2/xml/en/procedures/377/377-006-003.html>)

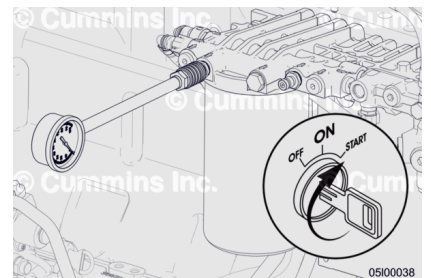


Measure the fuel inlet restriction. Refer to Procedure 006-020 in Section 6. (/qs3/pubsys2/xml/en/procedures/377/377-006-020.html)



Fuel lift pump pressure test:

- Install a 0 to 207 kPa [0 to 30 psi] pressure gauge at the Compuchek™ fitting at the inlet to the fuel filter head.
- Turn the keyswitch to the ON position to activate the lift pump. Repeat this action, if necessary.



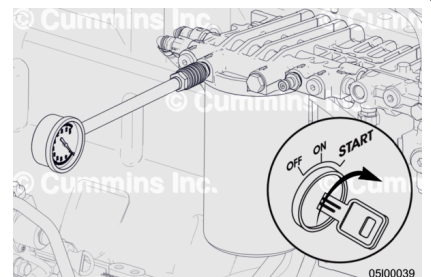
Measure the fuel lift pump pressure.

Measurements

	kpa	psi
Fuel Lift Pump Pressure (min)	69	10

Gear pump pressure test:

- Install a 0 to 207 kPa [0 to 30 psi] pressure gauge at the Compuchek™ fitting at the inlet to the fuel filter head.
- Disconnect the lift pump electrical connector, if equipped.
- Crank the engine and observe the fuel gear pump pressure.



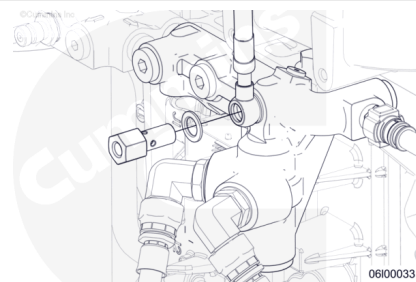
Measurements

	kpa	psi
Gear Pump Pressure While Cranking at a Minimum of 150 rpm for 10 seconds (min)	69	10

Fuel Drain Line Restriction

⚠ CAUTION ⚠

Hold banjo fittings while tightening the banjo bolt to prevent fitting rotation. Allowing the banjo fitting to rotate may damage the fuel line.

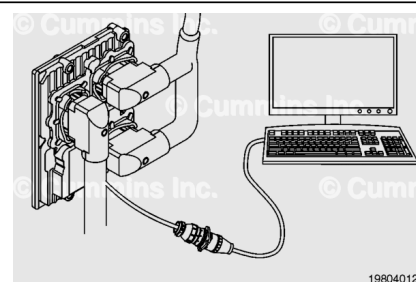


Measure the fuel drain line restriction. Refer to Procedure 006-012 in Section 6. (</qs3/pubsys2/xml/en/procedures/377/377-006-012.html>)

High-Pressure System Leak Down Test

Note : This test can not be performed if the engine will not start.

- Connect INSITE™ electronic service tool.
- Operate the engine.
- Perform INSITE™ electronic service tool Fuel System Leakage Test.
- Monitor the fuel rail pressure. Fuel rail pressure will rise and stabilize. This could take up to 5 minutes.
- Shut the engine OFF and wait for it to completely stop.
- Turn the keyswitch ON quickly.
- Record the Fuel Rail Pressure Measured value.
- Monitor for exactly 30 seconds.
- Record the Fuel Rail Pressure Measured value.
- Subtract the second measured value from the first measured value.
 - A decay in fuel pressure greater than 1500 bar [21756 psi] in 30 seconds is an indication of a high-pressure fuel system leak.



- Reference the appropriate troubleshooting symptom tree for repair direction.

High-Pressure Fuel Pump Performance Test

Initial Setup

Injector return fuel can be measured to determine if any injector leakage exists. Measurement of fuel injector leakage requires the use of various flow adapter hoses from the Fuel System Leak Test Kit, Part Number 4918354. These tools are used to isolate the leakage from just the injectors, so it can be measured with a graduated cylinder

If the low pressure fuel system is degraded or the fuel pressure relief valve leakage is out of specification, this test will **not** provide accurate results.

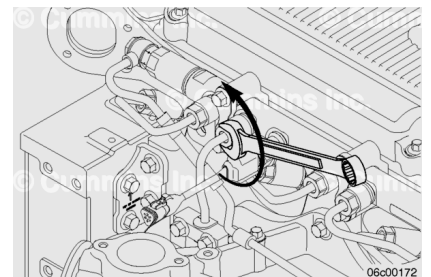
Troubleshoot the low-pressure system and any high-pressure fuel system leakage before performing this test.

⚠ WARNING ⚠

High-pressure fuel spray can cause serious injury or death.

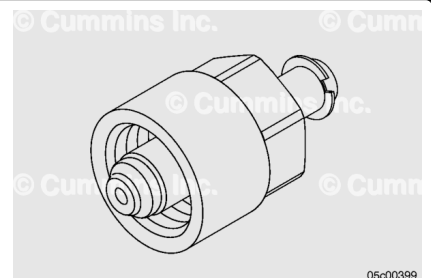
Relieve the high-pressure fuel from the fuel rail. Refer to Procedure 006-081 in Section 6.

(/qs3/pubsys2/xml/en/procedures/377/377-006-081.html)



⚠ WARNING ⚠

When servicing the engine, do not use the starting motor to rotate the engine with a high-pressure fuel system joint open. Rotating the engine can create highly pressurized fuel in the fuel system. High-pressure fuel spray can penetrate the skin, resulting in serious personal injury or death.



⚠ CAUTION ⚠

Do not install the isolation tool at the high-pressure pump outlet fitting. Severe engine damage will result. This tool must only be installed at the fuel rail for the purpose of isolating the high-pressure fuel supply from individual injectors.

This test requires five leak test isolation tools, Part Number 4919546.

Block off five injectors at the fuel rail in place of the high-pressure fuel line. Keep the injector line supply port that is most accessible unblocked.

Torque Value: 47 n•m [35 ft-lb]

Before installation, use electrical contact cleaner, Part Number 3824510, or equivalent, to thoroughly flush any residual fuel and/or debris from the high-pressure fuel pump service tool, Part Number 5299753.

Install the high-pressure fuel pump service tool, Part Number 5299753, from kit, 5299721, at the unblocked fuel rail connection.

Torque Value: 47 n•m [35 ft-lb]

Note : Cranking speed must be above 130 rpm. If necessary, use a battery charger to maintain proper cranking speed.

Note : Units equipped with starter lockout features may require additional cranking aids.

Minimum Cranking Speed: 130 rpm.

Measurement

Note : Do not crank the engine for 30 seconds continuously. Crank the engine in intervals with a 15 second pause between intervals. This reduces the possibility of overheating the starting motor.

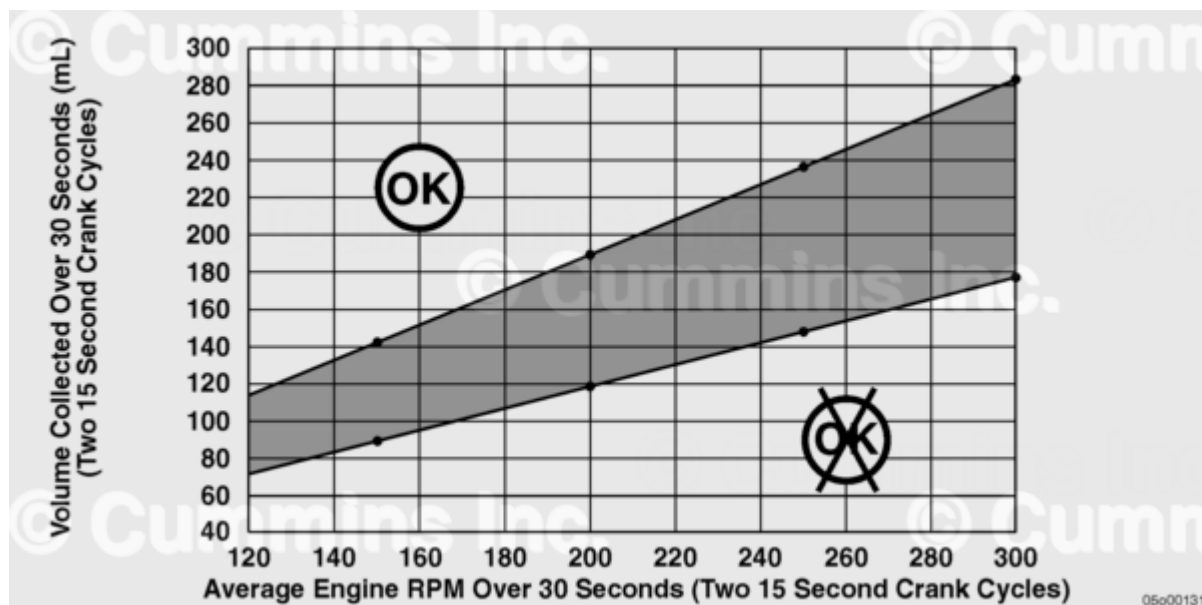
- With the following conditions met:
 - Connect fuel drain line, Part Number 3164673, to high-pressure fuel pump service tool, Part Number 5299753.
 - Disconnect the fuel pump actuator from the electrical harness.
 - Disconnect the injector pass-through harness from the engine wiring harness.
 - Route the hose, Part Number 3164673, into a graduated beaker, Part Number 4919139, for measurement.
 - Connect INSITE™ electronic service tool.
 - Log engine speed and fuel rail pressure with INSITE™ electronic service tool.
- Perform initial air purge - no measurements required.
- Crank until fuel flow exits the hose making sure air has been purged or for a maximum of 15 seconds.
- If no fuel flow exists after three 15 second crank cycles, measure the fuel pressure relief valve return flow while performing the High-Pressure Fuel Pump Performance Test. Follow the Fuel Pump Pressure Relief Performance Test in this procedure.

- If the fuel pressure relief valve return flow is out of specification, high-pressure fuel pump service tool, Part Number 5299753, is malfunctioning.
 - Remove the service tool and use electrical contact cleaner, Part Number 3824510, or equivalent, to thoroughly flush any residual fuel and/or debris from the high-pressure fuel pump service tool.
 - Install the high-pressure fuel pump service tool and crank until the fuel exits the hose or for maximum of 15 seconds.
 - If no fuel exits the hose after three 15 seconds crank cycles, high-pressure service tool, Part Number 5299753, is **not** working properly and **must** be replaced.
- If the fuel pressure relief valve return flow is in specification, inspect the fuel pump head. Refer to Procedure 005-227 in Section 5.
(/qs3/pubsys2/xml/en/procedures/377/377-005-227.html)
- Discard purged fuel from service tool hose. Dispose of in accordance with local and environmental regulations.

30 Second Fuel Volume Measurement - Two 15 Second Crank Cycles (Measurement begins when the fuel pump performance test service tool opens. A pressure pulse can be heard and felt through the high-pressure fuel pump service tool.)

- Collect the fuel that exits the high-pressure fuel pump service tool during the initial 15 second measurement cycle.
- Allow the starting motor to cool for 15 seconds.
- Repeat a 15 second measurement cycle.
- Add the initial and final 15 second fuel volume measurements exiting the high-pressure fuel pump service tool. (30 second volume total.)
- Record the average engine cranking speed from the INSITE™ electronic service tool logging results.

Note : After completing the test, fuel may still be contained in the drain line. Make sure to empty the line into the measuring container to obtain a proper measurement.



Flow from the service tool is directly related to engine speed.

Use the graph to determine the minimum flow from the service tool.

- Engine speed can be found on the horizontal axis.
- Service tool flow in milliliters can be found on the vertical axis.

If the measured fuel flow from the high-pressure fuel pump service tool at the recorded engine speed is in the upper region of the graph, the fuel pump is functioning properly.

If the measured fuel flow from the high-pressure fuel pump service tool at the recorded engine speed is in the lower region of the graph, the fuel pump is **not** functioning properly and **must** be replaced.

If the measured flow is in the gray area of the graph:

- Check to make sure the correct specifications are being used.
- Verify proper high-pressure fuel pump service tool installation.
- Make sure there are **not** external leaks.
- Verify all low-pressure system checks were made. Perform the High-Pressure Fuel Pump Performance Test again.

If the test readings are still in the gray area, the fuel pump head **must** be removed and the fuel pump camshaft and fuel pump camshaft tappet rollers **must** be inspected. If the fuel pump passes the inspection, contact technical support.

- Remove the test equipment.
- Install the fuel injector supply lines. Refer to Procedure 006-060 in Section 6. (</qs3/pubsys2/xml/en/procedures/377/377-006-060.html>)
- Connect the fuel pump actuator to the electrical harness.
- Connect the injector pass-through harness to the engine wiring harness.

Note : Do not disassemble the service tool, Part Number 5299753.

High-Pressure Fuel Pump Return Flow Test

Initial Setup

⚠ CAUTION ⚠

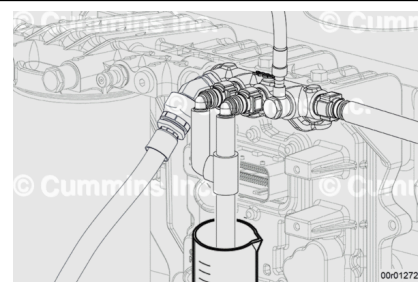
Installation of the banjo flow adapter at any place other than the recommended locations can cause damage to high-pressure fuel system components.

Return fuel is routed from the injectors, the fuel rail pressure relief valve, and the fuel pump head through three different drain lines. Measurement of High-Pressure Fuel Pump Return Flow requires the use of a fuel return hose, included in tester, fuel system leak, Part Number 4918354.

Obtain a graduated cylinder that is marked in cubic centimeters. Use graduated cylinder, Cummins® Part Number 4919139, or equivalent.



- Disconnect the high-pressure fuel pump drain line at the fuel drain manifold.
- From the Fuel System Leak Test Kit, Part Number 4918354, connect a female extension line to the fuel drain manifold. Route this line into a collection device (bucket).
- From Fuel System Leak Test Kit, Part Number 4918354, connect a male extension hose to the quick disconnect connection of the high-pressure pump drain hose.



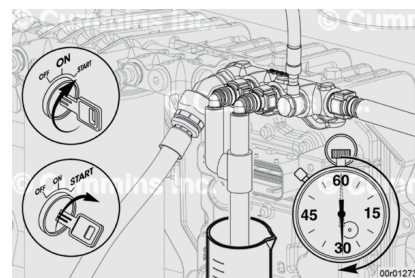
Route the hose into a collection device.

Use graduated cylinder, Cummins® Part Number 4919139, for measurement when directed.

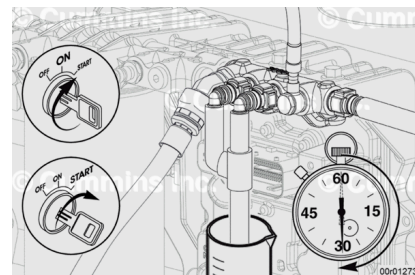
Measurement - Engine Will Start

- Close the engine cover while performing this measurement.
- Operate the engine until it is at operating temperature.

- Allow the engine to idle with fuel flowing into a collection device (bucket).
- Perform INSITE™ electronic service tool Fuel System Leakage Test.
- Monitor the fuel pressure rise on the screen.
- Allow the pressure to stabilize and create a steady stream of flow. This could take up to 5 minutes.



- Move the flow adapter line from the bucket to the graduated cylinder, Cummins® Part Number 4919139.
- Allow fuel to flow into the graduated cylinder for exactly 30 seconds.
- Return the flow adapter line back to the collection device (bucket) immediately.
- Record the volume of fuel collected in the graduated cylinder. Compare it to the specification below.



Maximum Fuel Volume Measurement During High Pressure Leakage Test

Time	ml/cc or fl-oz
30 Seconds	525 or 18

If 525 ml/cc [18 fl-oz] pump head drain flow is collected in less than 30 seconds, the fuel pump head has malfunctioned and **must** be replaced.

Minimum Fuel Volume Measurement During High Pressure Leakage Test

Time	ml/cc or fl-oz
30 Seconds	150 or 5

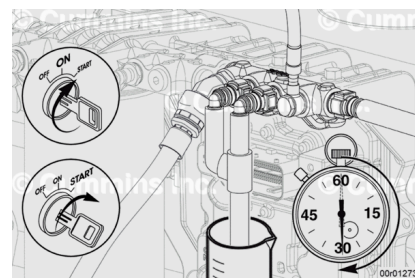
If less than 150 ml/cc [5 fl-oz] pump head drain flow is collected in 30 seconds, the fuel pump adapter/inlet metering valve has malfunctioned and **must** be replaced.

Note : This specification is valid for engines operating on diesel fuels. Low fuel viscosity will increase the leakage rate; for example, kerosene or aviation fuels will result in excessive leakage. Verify the fuel type being measured before replacing a fuel pump head for excessive leakage.

Measurement - Engine Will Not Start

Note : Do not crank the engine for 30 seconds continuously. Crank the engine in 15 second intervals with a 15 second pause between intervals. This reduces the possibility of overheating the starting motor.

- Connect the fuel pump diagnostic line at the fuel drain manifold.
- Disconnect the lift pump.
- Route the hose into a collection device (bucket).
- Use graduated cylinder, Cummins® Part Number 4919139, for measurement when directed.
- Close the engine cover while performing this measurement.
- Begin cranking the engine.
- Obtain a steady stream of fuel flow from the drain line.
- Move the drain line to the graduated cylinder for 15 seconds then back to the bucket.
- Stop cranking and allow the engine to rest for 15 seconds.
- Repeat the three steps above a total of two times.
- Record the amount of fuel flow from the injector drain line in 30 seconds while the engine is cranking.
- Record the volume of fuel collected in the graduated cylinder. Compare it to the specification below.



Maximum Fuel Volume Measurement During High Pressure Leakage Test

Time	ml/cc or fl-oz
30 Seconds	390 or 13

If more than 390 ml/cc [13 fl-oz] pump head drain flow is collected in less than 30 seconds, the fuel pump head has malfunctioned and **must** be replaced.

Minimum Fuel Volume Measurement During High Pressure Leakage Test

Time	ml/cc or fl-oz
30 Seconds	150 or 5

If less than 150 ml/cc [5 fl-oz] pump head drain flow is collected in 30 seconds, the fuel pump adapter/inlet metering valve has malfunctioned and **must** be replaced.

Note : This specification is valid for engines operating on diesel fuels. Low fuel viscosity will increase the

leakage rate; for example, kerosene or aviation fuels will result in excessive leakage. Verify the fuel type being measured before replacing a fuel pump head for excessive leakage.

High-Pressure Injector Return Flow Test

Initial Setup

⚠ CAUTION ⚠

Installation of the banjo flow adapter at any place other than the recommended locations can cause damage to high-pressure fuel system components.

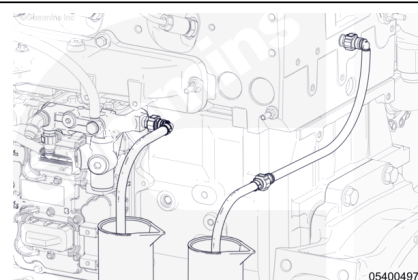


Injector return fuel can be measured to determine if any injector leakage exists. Measurement of fuel injector leakage requires the use of various flow adapter hoses from Fuel System Leak Test Kit, Part Number 4918354. These tools are used to isolate the leakage from just the injectors, so it can be measured with a graduated cylinder.

Obtain a graduated cylinder that is marked in cubic centimeters. Use graduated cylinder, Part Number 4919139, or equivalent.

⚠ CAUTION ⚠

Do not use a screwdriver or pliers on the low-pressure fuel line locking tabs. This may result in damage to the quick disconnect fittings.



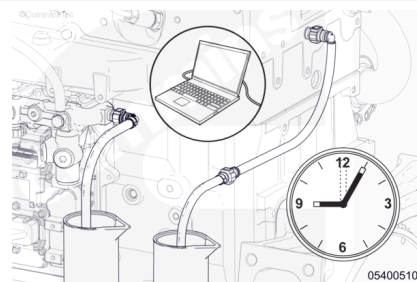
Note : Make sure the engine is at operating temperature before beginning this test.

- Disconnect the injector drain line at the fuel drain manifold.
- From the Fuel System Leak Test Kit, Part Number 4918354, connect a female extension line to the fuel drain manifold. Route this line into a collection device (bucket).
- From Fuel System Leak Test Kit, Part Number 4918354, connect a male extension hose to the quick disconnect connection of the injector drain hose.
- Route the hose into a collection device (bucket).

- Use graduated cylinder, Part Number 4919139, for measurement when directed.

Measurement - Engine Will Start

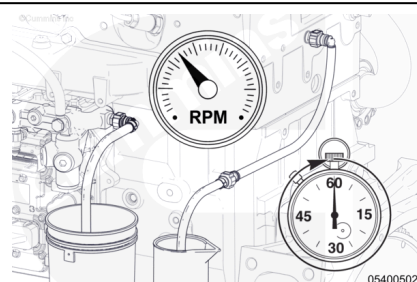
- Close the engine cover while performing this measurement.
- Connect INSITE™ electronic service tool.
- Go to INSITE™ electronic service tool High-Pressure Leakage Test.
- Start the engine and allow it to idle with fuel flowing into a bucket.
- Wait for the lift pump cycle to complete (about 60 seconds).
- Run the engine until it reaches operating temperature.
- Begin the High-Pressure Leakage Test.



Note : Make sure a steady flow of fuel is present at the drain line before beginning the measurement. Air in the line and movement of the hose during measurement can result in inaccurate measurements.

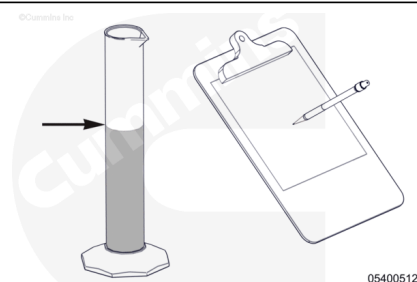
- Monitor the fuel pressure rise on the screen.
- Allow the pressure to stabilize and create a steady stream of flow. This could take up to 5 minutes.
- When fuel pressure stabilizes, allow fuel to flow into the collection device for 60 seconds.

- Move the flow adapter line from the bucket to the graduated cylinder, Part Number 4919139.
- Allow fuel to flow into the graduated cylinder for exactly 60 seconds.
- Immediately return the flow adapter line back to the collection device (bucket).



- Record the volume of fuel collected in the graduated cylinder. Compare it to the specification below.

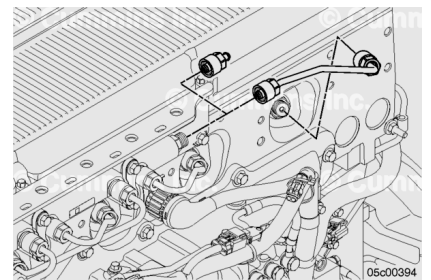
Maximum Injector Return Flow Measurement	
Elapsed Time	ml/cc or fl-oz
60 Seconds	200 or 7



Note : Fuel temperature and fuel type will influence this measurement. For example, as the engine is warmed up and the injectors become hot, the leakage rate will increase. Also, low viscosity fuels, such as kerosene or aviation fuels, will cause the leakage rate to increase. The above specification is correct for on-highway diesel fuels, where the diesel fuel inlet temperature is less than 49°C [120°F].

- After recording the fuel leakage quantity, record the Fuel Rail Pressure Measured value.
- Stop the INSITE™ electronic service tool High-Pressure Leak Test. Turn the keyswitch to OFF.

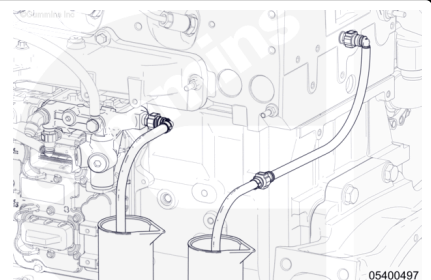
If injector drain flow is excessive, it will be necessary to isolate the injector(s) or fuel connector(s) that are damaged or worn.



Measurement - Engine Will Not Start

Note : Do not crank the engine for 30 seconds continuously. Crank the engine in 15 second intervals with a 15 second pause between intervals. This reduces the possibility of overheating the starting motor.

- Lift pump disconnected.
- Route the hose into a collection device (bucket).
- Use graduated cylinder, Part Number 4919139, for measurement when directed.
- Close the engine cover while performing this measurement.

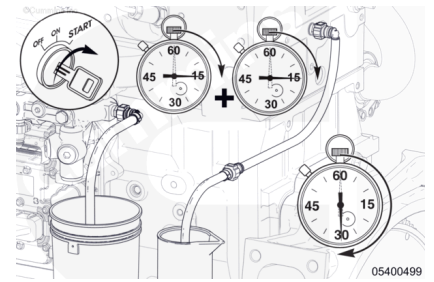


The leakage should **only** be a few drops.

If injector drain flow is excessive, it will be necessary to isolate the injector(s) or fuel connector(s) that are damaged or worn.

- Begin cranking the engine.
- Begin cranking the engine until fuel exits the drain line.
- Move the drain line to the graduated cylinder for 15 seconds, then back to the bucket.
- Stop cranking. Allow the engine to rest for 15 seconds.
- Repeat the three steps above.

- Record the amount of fuel flow from the injector drain line in 30 seconds while the engine is cranking.
- The leakage should **only** be a few drops.
- If the injector drain flow is excessive, it will be necessary to isolate the injector(s) or fuel connector(s) that are damaged or worn.



High-Pressure Injector Return Flow Isolation Test

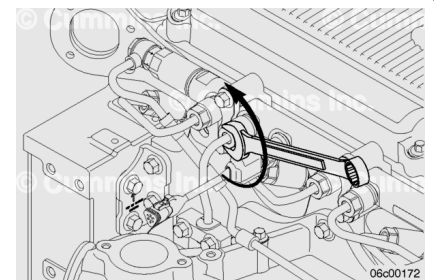
Initial Setup

⚠ WARNING ⚠

High-pressure fuel spray can cause serious injury or death.

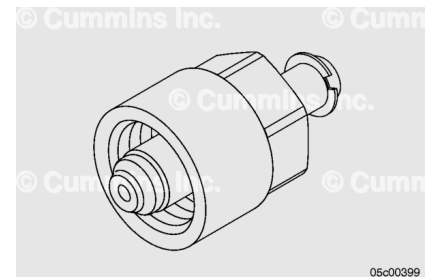
Relieve the high-pressure fuel from the fuel rail. Refer to Procedure 006-081 in Section 6.

(/qs3/pubsys2/xml/en/procedures/377/377-006-081.html)



⚠ WARNING ⚠

When servicing the engine, do not use the starting motor to rotate the engine with a high-pressure fuel system joint open. Rotating the engine can create highly pressurized fuel in the fuel system. High-pressure fuel spray can penetrate the skin, resulting in serious personal injury or death.



⚠ CAUTION ⚠

Do not install the isolation tool at the high-pressure pump outlet fitting. Severe engine damage will result. This tool must only be installed at the fuel rail for the purpose of isolating the high-pressure fuel supply from individual injectors.

⚠ CAUTION ⚠

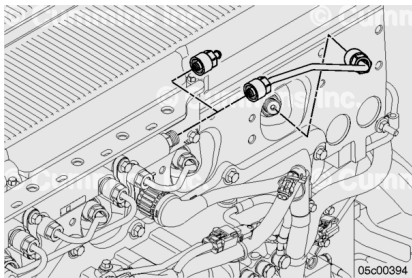
Make sure the keyswitch is in the OFF position (engine not operating) when loosening or tightening high-pressure fuel lines.

Use leak test isolation tool, Part Number 4919546, to isolate excessive fuel drain from injectors or fuel connectors.

Relieve the high-pressure fuel from the fuel rail. Refer to Procedure 006-081 in Section 6.
(/qs3/pubsys2/xml/en/procedures/377/377-006-081.html)

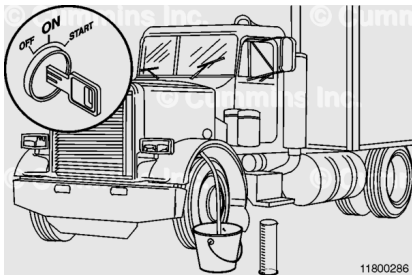
Isolate the injector and fuel connector for each cylinder by installing the isolation tool at the fuel rail in place of the high-pressure fuel line that supplies the fuel connector.

Torque Value: 47 n•m [35 ft-lb]



Measurement - Engine Will Start

- Run engine at operating temperature.
- Disconnect the lift pump.
- Connect injector drain line at the fuel drain manifold.
- Route the hose into a collection device (bucket).
- Use graduated cylinder, Part Number 4919139, for measurement when directed.
- Close the engine cover while performing this measurement.
- Connect INSITE™ electronic service tool.
- Start the engine and allow it to idle with fuel flowing into a collection device (bucket).
- Begin the High-Pressure Leakage Test.
- Monitor the fuel pressure rise on the screen.
- Allow the pressure to stabilize and create a steady stream of flow. This could take up to 5 minutes.



- Record the Fuel Rail Pressure Measure value.
- Move the banjo flow adapter line from the bucket to the graduated cylinder, Part Number 4919139.
- Allow fuel to flow into the graduated cylinder for exactly 60 seconds.
- Return the banjo flow adapter line back to the collection device (bucket) immediately.
- Record the quantity of fuel collected in the graduated cylinder. Compare it to the specification below.

Do this up to six times, once while each injector is isolated.

Cylinder	Pressure	Quantity
1		
2		
3		
4		
5		
6		

Note : Make sure a steady flow of fuel is present at the drain line before beginning the measurement. Air in the line and movement of the hose during measurement can result in inaccurate measurements.

Maximum Injector Return Flow Measurement	
Elapsed Time	ml/cc or fl-oz
60 Seconds	200 or 7

If isolating any single injector results in a significant increase in fuel rail pressure measured, or a reduction in the maximum injector return flow measurement:

- Check or tighten that fuel connector. Refer to Procedure 006-052 in Section 6.
(/qs3/pubsys2/xml/en/procedures/377/377-006-052.html)
- Remove the isolation tool.
- Run the High-Pressure Injector Return Flow Test.

If a significant increase in Fuel Rail Pressure Measured, or the maximum injector return flow measurement is now within specification, the leak was caused by a leaking fuel connector.

If the leak remains, the fuel injector should be replaced.

Complete this up to six times, once while each injector is isolated.

Note : Make sure a steady flow of fuel is present at the drain line before beginning the measurement. Air in the line and movement of the hose during measurement can result in inaccurate measurements.

Measurement - Engine Will Not Start

Note : Do not crank the engine for 30 seconds continuously. Crank the engine in 15 second intervals with a 15 second pause between intervals. This reduces the possibility of overheating the starting motor.

- Injector drain line connected at the fuel drain manifold.
- Lift pump disconnected.
- Route the hose into a collection device (bucket).
- Use graduated cylinder, Part Number 4919139, for measurement when directed.
- Close the engine cover while performing this measurement.

Cylinder	Pressure	Quantity
1		
2		
3		
4		
5		
6		

05p00016

- Begin cranking the engine.
- Obtain a steady stream of fuel flow from the drain line.
- Move the drain line to the graduated cylinder for 15 seconds then back to the bucket.
- Stop cranking and allow the engine to rest for 15 seconds.
- Repeat the three steps above one more time.
- Record the amount of fuel flow from the injector drain line in 30 seconds while the engine is cranking.

If isolating any single injector results in a significant increase in Fuel Rail Pressure Measured or a reduction in the return flow measurement:

- Check or tighten that fuel connector. Refer to Procedure 006-052 in Section 6.
(/qs3/pubsys2/xml/en/procedures/377/377-006-052.html)
- Remove the isolation tool.
- Run the High-Pressure Injector Return Flow Test.

If a significant increase in Fuel Rail Pressure Measured or the leakage is now **only** a few drops, the leak was caused by a leaking fuel connector.

If the leak remains, the fuel injector should be replaced.

Do this up to six times, once while each injector is isolated.

Note : Make sure a steady flow of fuel is present at the drain line before beginning the measurement. Air in the line and movement of the hose during measurement can result in inaccurate measurements.

Fuel Pressure Relief Valve Return Flow Test

Initial Setup

CAUTION

Installation of the banjo flow adapter at any place other than the recommended locations can cause damage to high-pressure fuel system components.

Return fuel is routed from the injectors, the fuel rail pressure relief valve, and the fuel pump head through three different drain lines. Measurement of fuel injector leakage requires the use of a fuel return hose, Part Number 5299319.

The tool is used to isolate the leakage from just the injectors, so it can be measured with a graduated cylinder.

Obtain a graduated cylinder that is marked in cubic centimeters. Use graduated cylinder, Part Number 4919139, or equivalent.

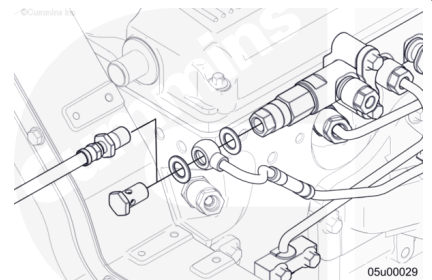


Remove the banjo bolt from the drain line at the fuel pressure relief valve.

Install the banjo flow adapter fitting, Part Number 5299319, at the fuel pressure relief valve.

Torque Value: 28.5 n•m [21 ft-lb]

Route the hose into a collection device.

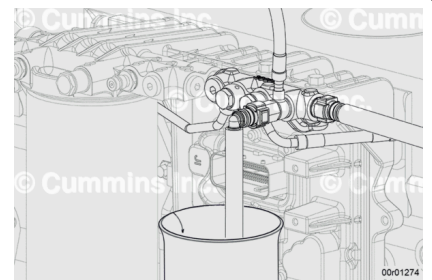


Alternatively, fuel return hose, Part Number 3164618, can be installed at the fuel drain manifold.

Torque Value: 12 n•m [105 in-lb]

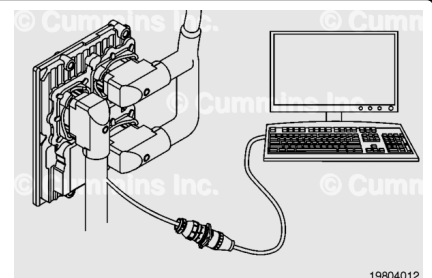
If the fuel rail return flow does **not** meet the specification when checking at the fuel drain manifold, measure the fuel rail return flow at the fuel rail pressure relief valve, **not** at the fuel drain manifold.

If the fuel rail return flow does **not** meet specification at the fuel rail pressure relief valve, a malfunctioning fuel rail relief valve has been detected.



Measurement - Engine Will Start

- Run engine at operating temperature.
- Connect the fuel pressure relief valve line at the fuel pressure relief valve.
- Route the hose into a collection device (bucket).
- Use graduated cylinder, Part Number 4919139, for measurement when directed.
- Close the engine cover while performing this measurement.
- Connect INSITE™ electronic service tool.



- Start the engine and allow it to idle with fuel pressure relief valve line in a collection device (bucket).
- Perform INSITE™ electronic service tool High-Pressure Leakage Test.
- Monitor the fuel pressure rise on the screen.
- Allow the pressure to stabilize. This could take up to 5 minutes.
- Move the fuel pressure relief valve line to a graduated cylinder, Part Number 4919139.
- Allow fuel to flow into the graduated cylinder for 60 seconds.

The leakage **must** be less than 10 drops per minute.

Replace the fuel pressure relief valve if leakage is greater than 10 drops in 60 seconds.

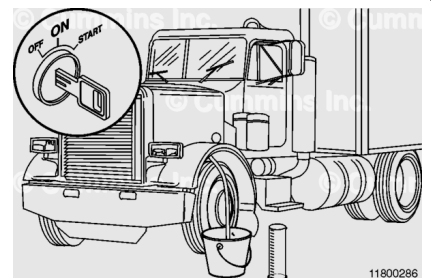
Note : If Fault Code 449 or 2311 is active, do not replace the fuel pressure relief valve without first determining the cause of the fault condition.

Note : A steady leak from the fuel pressure relief valve can be an indication that the valve seat is worn. This condition will often coincide with Fault Code 1911 and/or Fault Code 3727.

Measurement - Engine Will Not Start

Note : Do not crank the engine for 60 seconds continuously. Crank the engine in 15 second intervals with a 15 second pause between intervals. This reduces the possibility of overheating the starting motor.

- Connect the fuel pressure relief valve line at the fuel pressure relief valve.
- Disconnect the lift pump.
- Route the hose into a collection device (bucket).
- Use graduated cylinder, Part Number 4919139, for measurement when directed.
- Close the engine cover while performing this measurement.
- Begin cranking the engine.
- Move the drain line to the graduated cylinder for 15 seconds then back to the bucket.
- Stop cranking and allow the engine to rest for 15 seconds.
- Repeat the three steps above a total of four times.
- Record the amount of fuel flow from the fuel pressure relief valve line in 1 minute while the engine is cranking.



The leakage should be less than 10 drops per minute.

Replace the fuel pressure relief valve if leakage is greater than 10 drops in 60 seconds.

Note : If Fault Code 449 or 2311 is active, do not replace the fuel pressure relief valve without first determining the cause of the fault condition.

High-Pressure Fuel Pump Supply Flow Test

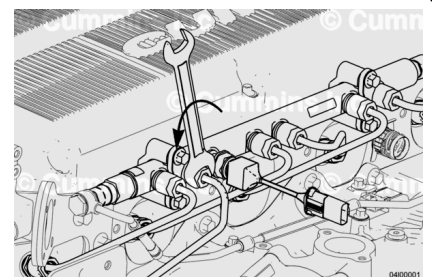
Initial Setup

Obtain a graduated cylinder marked in cubic centimeters or milliliters, graduated beaker, Part Number 4919139, or equivalent.



⚠ WARNING ⚠

Normal engine operation creates highly pressurized fuel in the fuel line which will remain in the fuel line after engine shutdown. Never open the fuel system when the engine is operating. Before servicing the fuel system, always loosen the pump to rail fuel line at the rail to vent the pressure. Keep hands clear of the line when loosening. High-pressure fuel spray can penetrate the skin, resulting in serious personal injury or death.



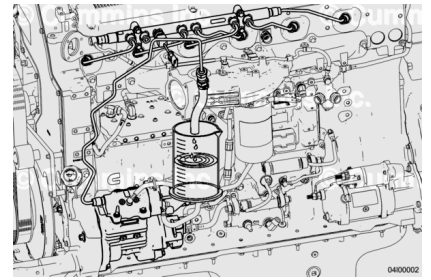
- Before servicing the fuel system, loosen the pump-to-rail line at the rail to vent the pressure. Keep hands clear of the line when loosening. Tighten the fuel rail nut once pressure has been released.

Torque Value: 47 N·m [35 ft-lb]

Note : A machined slot in this fitting directs the fuel spray toward the engine block.

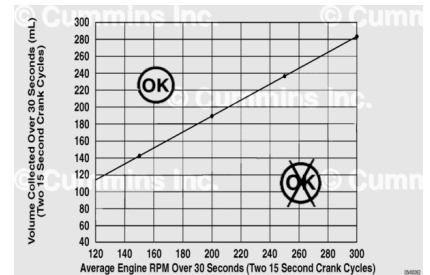
- Loosen the fuel pump-to-rail supply line clamps.
- Disconnect the high-pressure fuel line from the rail.
- Clamp a hose to the end of the fuel line.

- Route a hose from the test fitting into a clean graduated cylinder.
- Disconnect the fuel pump actuator from the electrical wiring harness.
- Verify all low-pressure system checks were made.



Measurement

- Crank the engine for 30 seconds. Make sure engine cranking speed is at least 150 rpm.
- If there is no flow, verify that the vehicle has been fully primed.



If the measured fuel flow from the high-pressure fuel pump at the recorded engine speed is in the upper region of the graph, the fuel pump is functioning properly.

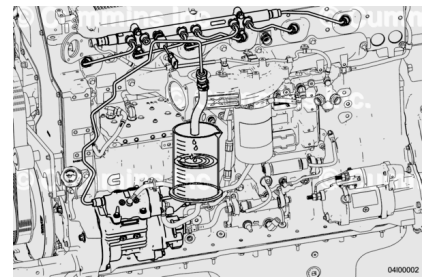
If the measured fuel flow from the high-pressure fuel pump at the recorded engine speed is in the lower region of the graph, the fuel pump head is malfunctioning.

The fuel pump head **must** be removed and the fuel pump camshaft and fuel pump camshaft tappet rollers **must** be inspected. Replace the fuel pump head. Refer to Procedure 005-227 in Section 5.

(/qs3/pubsys2/xml/en/procedures/377/377-005-227.html)

Note : Do not crank the engine for 30 seconds continuously. Crank the engine in 15 second intervals with a 15 second break between cranking. This reduces the possibility of overheating the starter motor.

Note : Some vehicles are equipped with an Engine Starting Motor Protection feature. If the starting motor is engaged for 30 or more seconds, without the engine starting, the starting motor will be locked out from operating, allowing for proper cooling of the starting motor. During this time the Wait to Start lamp, if equipped, will flash for 2 minutes. Once the lamp discontinues flashing, the starting motor will be allowed to function again.



Finishing Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect all diagnostic test fittings and install all components removed during testing. Refer to Procedure 006-024 (</qs3/pubsys2/xml/en/procedures/377/377-006-024.html>) in Section 6.
- Connect the batteries. See equipment manufacturer service information.
- Bleed air from the fuel system. Refer to Procedure 005-234 (</qs3/pubsys2/xml/en/procedures/377/377-005-234-tr.html>) in Section 5.
- Operate the engine. Check for leaks or performance complaints.